Karhunkierros SAR

The Complete Wilderness Rescue System

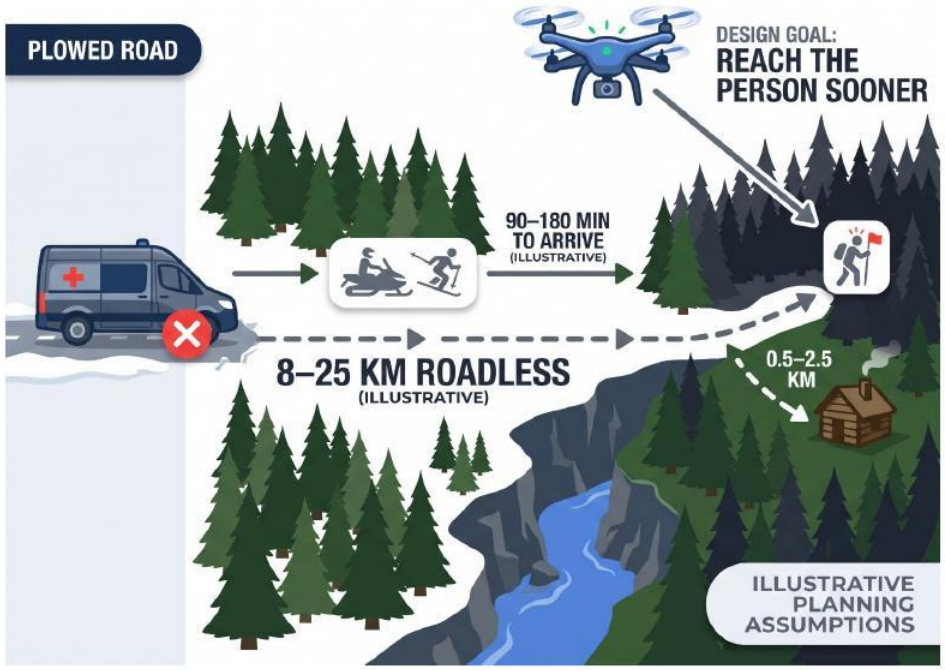
Civilian search and rescue for Finland's roadless wilderness

For technical partners and rescue stakeholders · 20 October 2026

Open-source design reference. Not a certified aircraft or rescue service.



The Problem: Help Can Be Hours Away, Shelter Is Often Closer



8–25 km

ILLUSTRATIVE DISTANCE, LOST HIKER TO PLOWED ROAD

90–180 min

ILLUSTRATIVE SNOWMOBILE OR SKI PATROL ARRIVAL

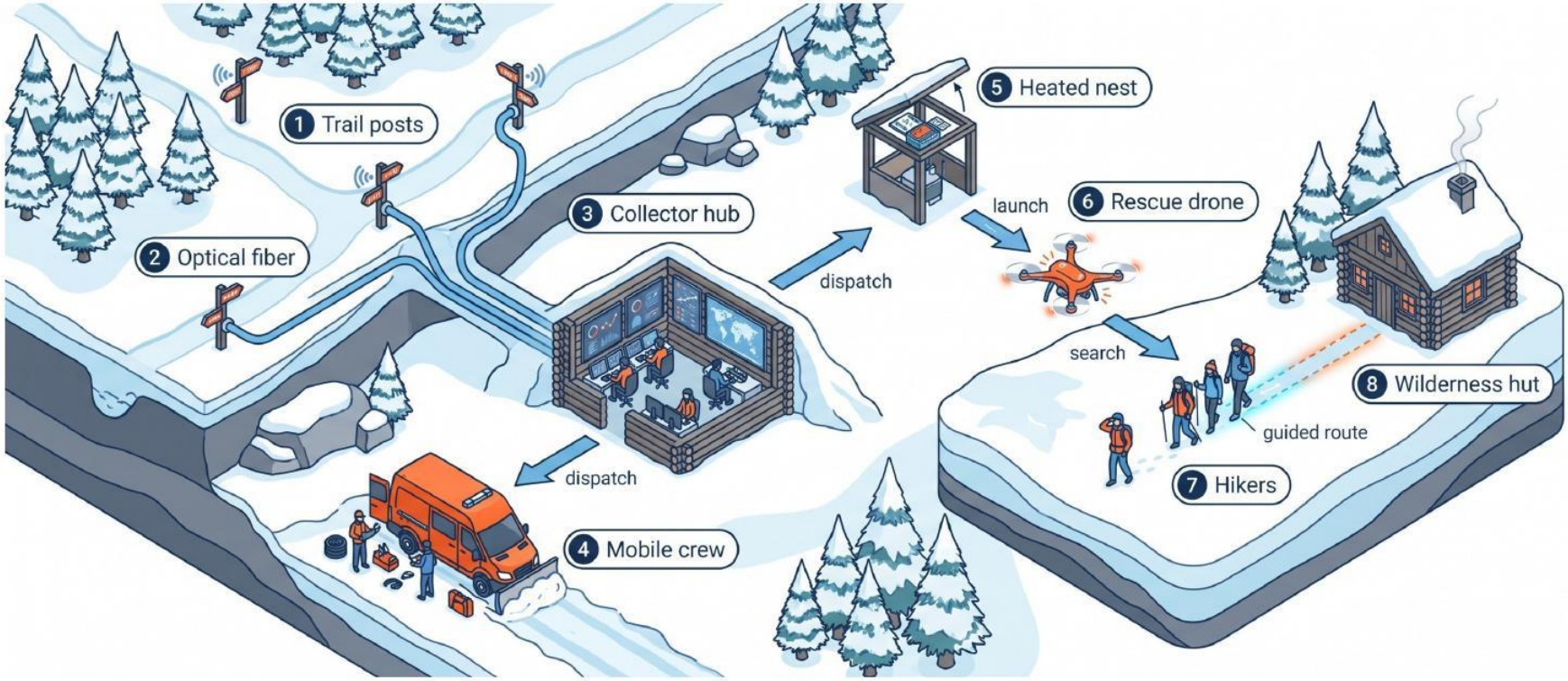
450+

WILDERNESS AND DAY HUTS (FACILITIES VARY BY HUT)

Mission: find the person fast, keep them warm, and guide them to the nearest hut, or keep watch until the ground team arrives.

8–25 km and 90–180 min are illustrative planning assumptions from the project packet, not agency statistics. Hut stoves and supplies are not guaranteed at every hut. Drone timings in this deck are design targets.

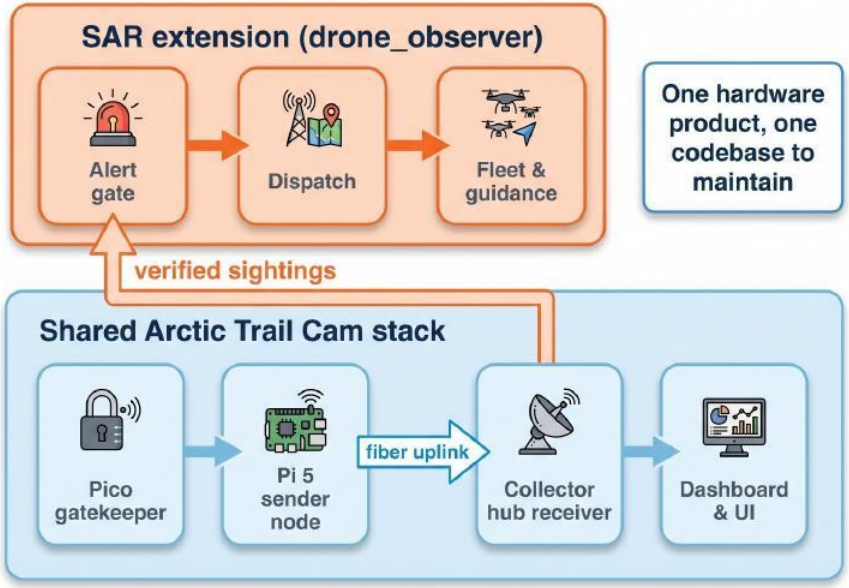
One System, Eight Parts Working Across the Landscape



The Rescue in Five Stages



One Trail Camera, Two Missions



Wildlife trails and rescue share one stack

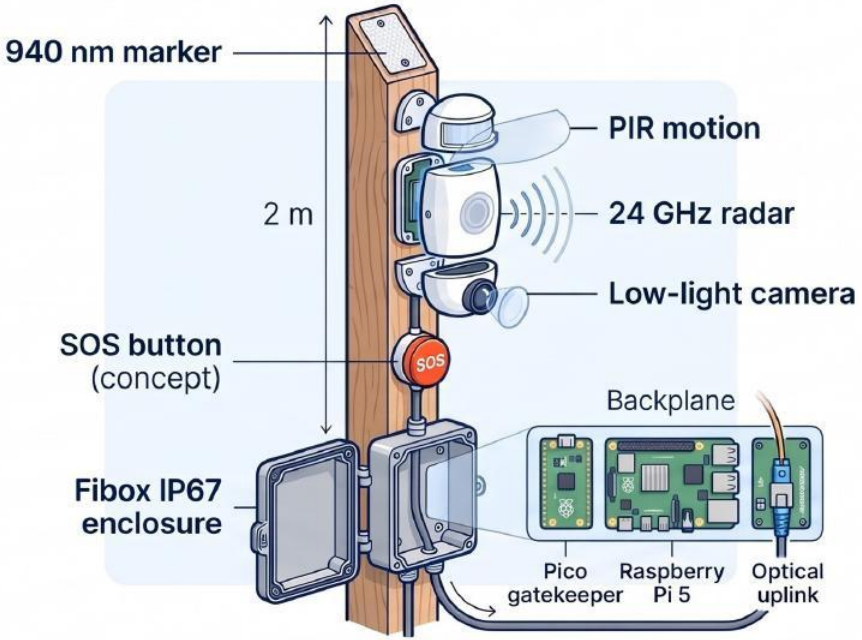
Ground node and collector hub are **100% shared** with the arctic-trail-cam project

- Only one trail camera to build and maintain
- SAR logic lives in a separate module and repository

Off-the-shelf snap-in parts, **no custom circuit boards**

- MIT-licensed reference design

Inside the Trail Post



Data leaves by fiber, not radio. The radar still emits.

Sightings Expire After 48 h Unless an Alert Unlocks Them



What is stored

- Group size, time, speed, heading, coarse jacket colour

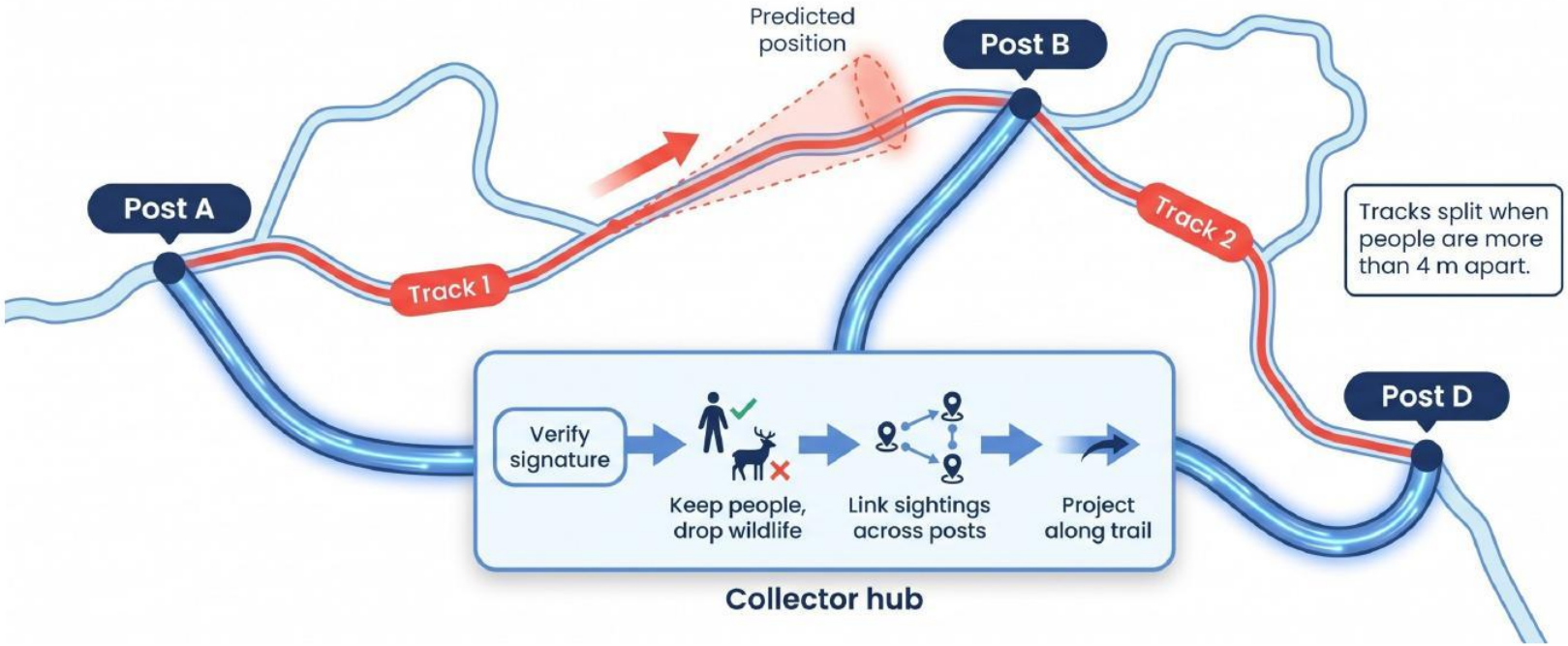
Kept on the post, deleted after **172,800 s**

What is never done

- No facial recognition
- No cloud upload
- No following routine hikers

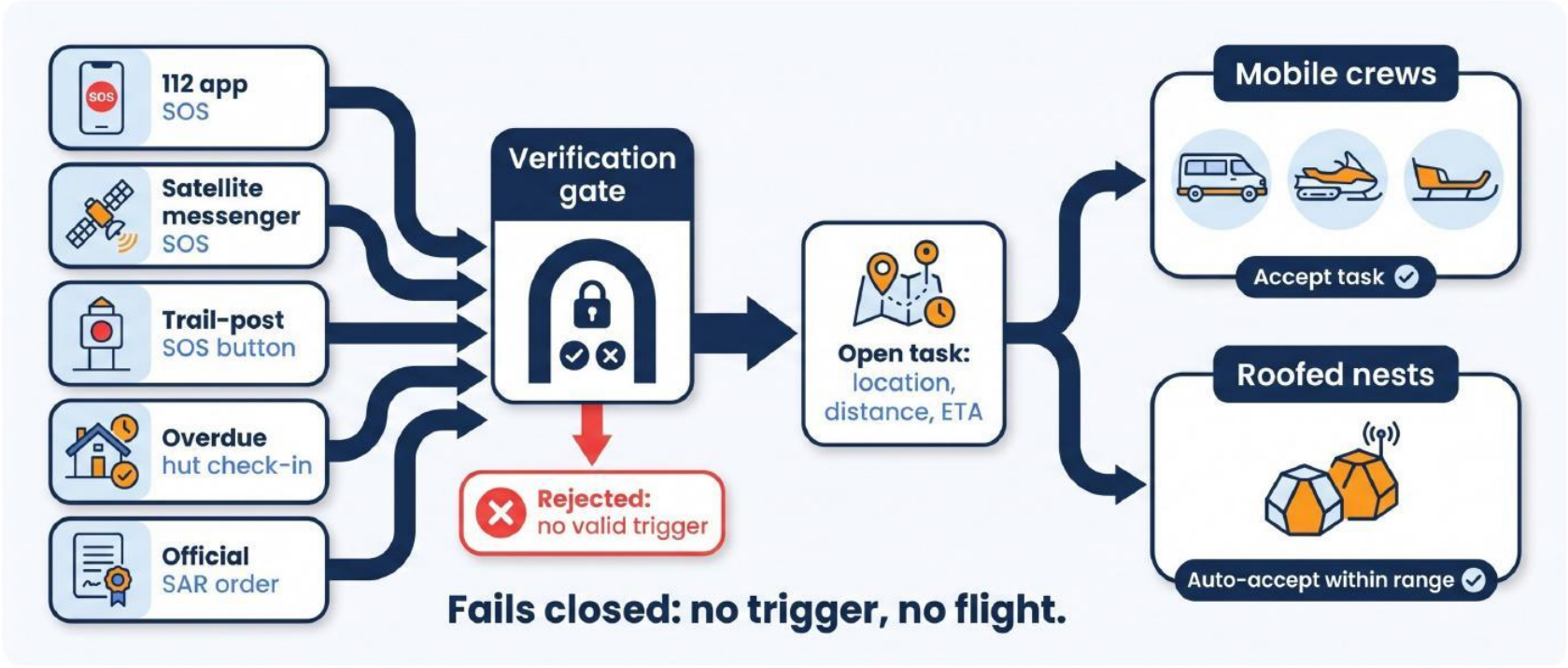
Qualification: these are design controls, not a legal-compliance claim. Anonymous tracks can still be privacy-sensitive.

The Collector Hub Fuses Sightings into Tracks



Fusion pipeline verified in simulation; posts send data to the hub over fiber, not radio.

A Verified Alert Becomes an Open Task for Every Nearby Unit

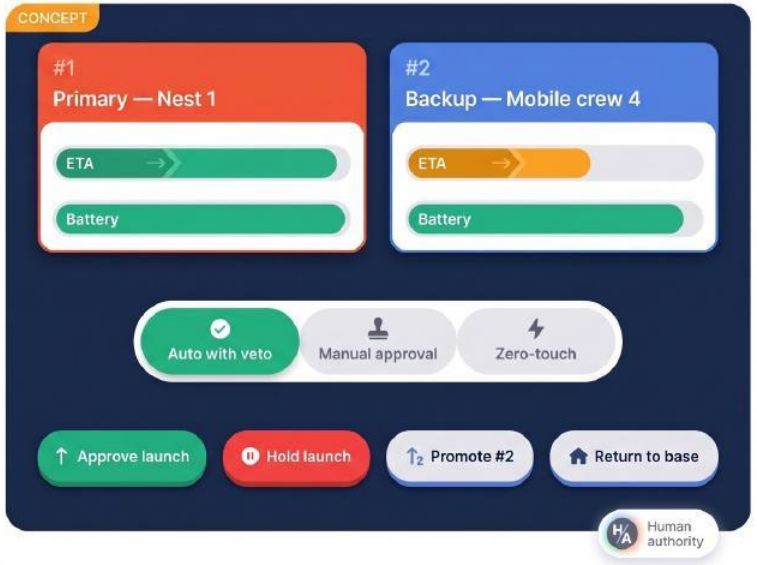


112, satellite-messenger and agency links are proposed interfaces, not live integrations or partnerships. Gate logic is simulated.

The System Ranks Volunteers; a Human Keeps Authority

Default scoring of each accepting unit

Factor	Weight
Flight ETA	45%
Winter battery margin	25%
Sensor and payload capability	15%
Readiness of shelter and links	15%



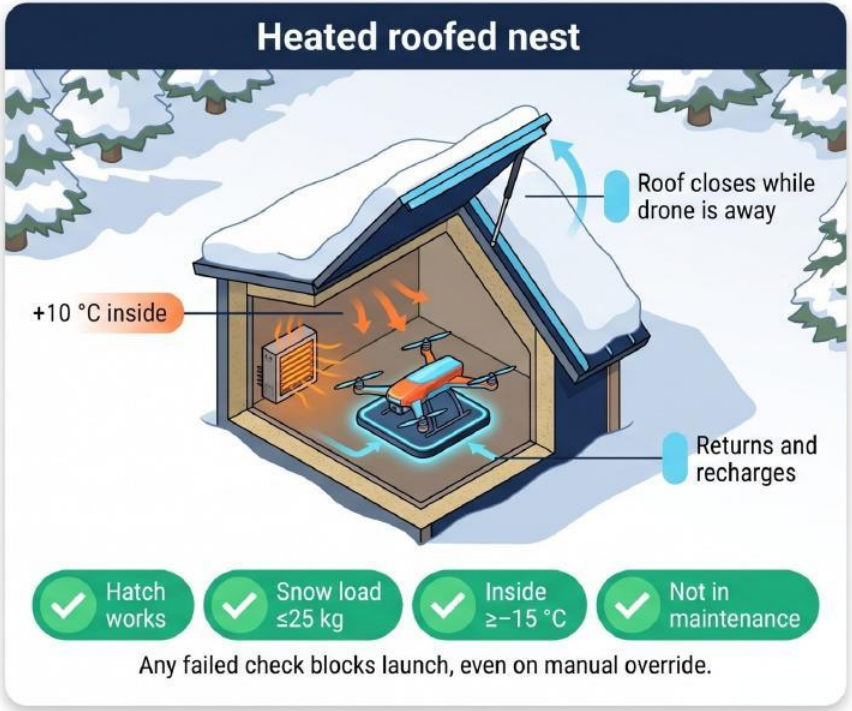
#1 Primary rescuer and **#2 Backup** are awarded

- If #1 aborts, #2 is re-checked live before promotion

No feasible, safe unit means **no launch**

Two Ways to Launch: Mobile Crew or Heated Roofed Nest





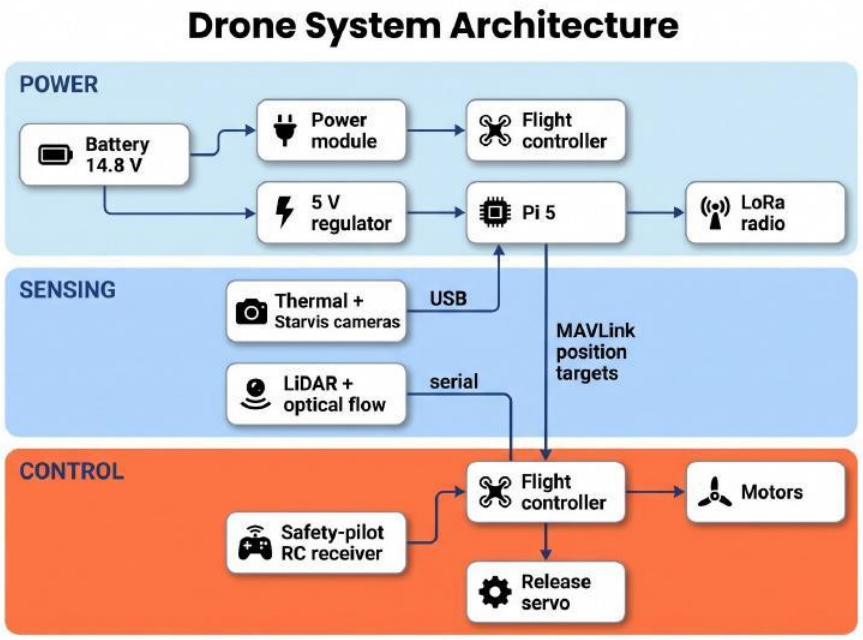
Interlock thresholds and nest temperature are design values; the 14 km auto-accept radius is a configurable default.

The Rescue Aircraft, Part by Part



Concept catalog of reference COTS parts (v2.0 BOM). Compatibility and integration are still to be verified.

Power, Data and Control Flow



Who does what in the aircraft

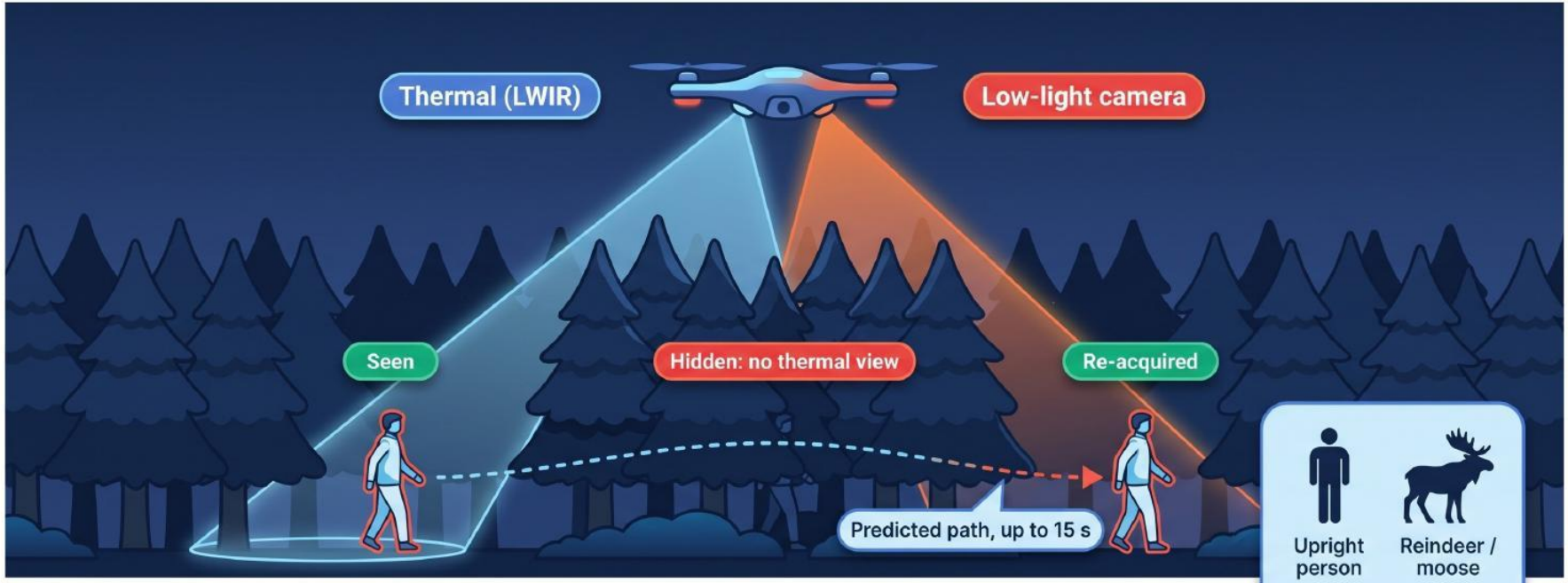
Pixhawk 6C (ArduCopter 4.5+): stabilises, fuses navigation, runs failsafes

Raspberry Pi 5: detects people, tracks, plans guidance, enforces the 15 m standoff (design constraint)

Safety pilot: can take over instantly with LOITER, GUIDED or RTL

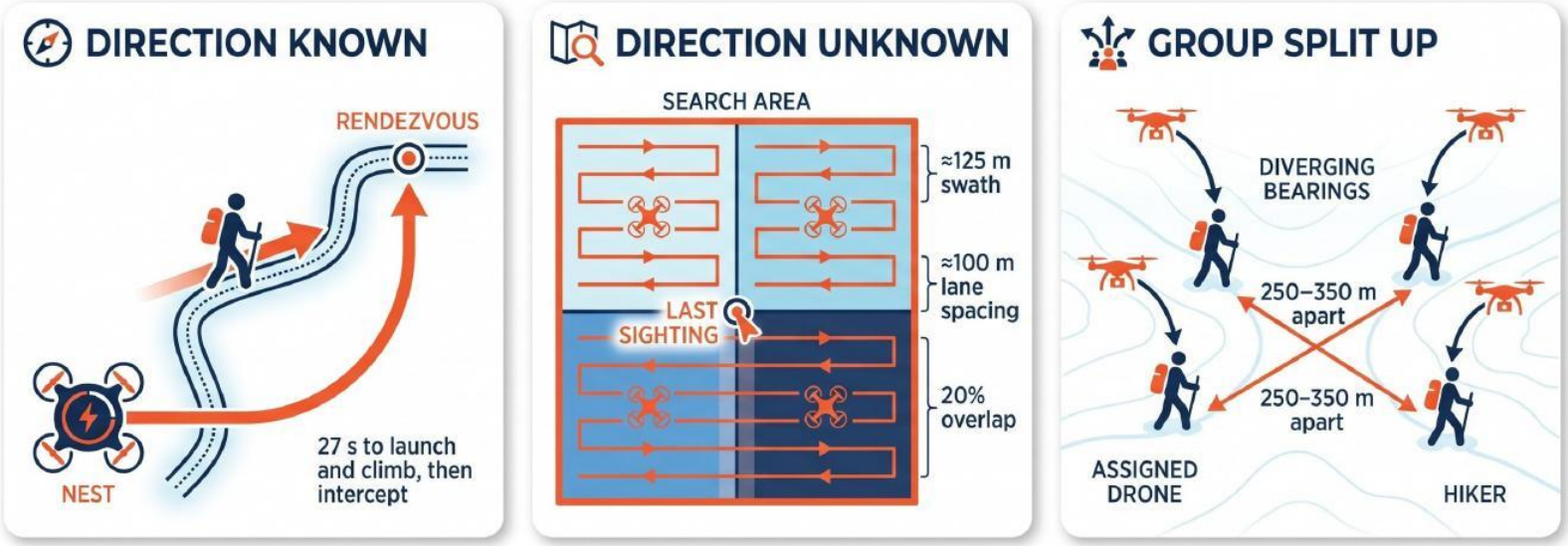
Coordinator: approves, holds or recalls

Seeing People at Night, and the Limits Under Canopy



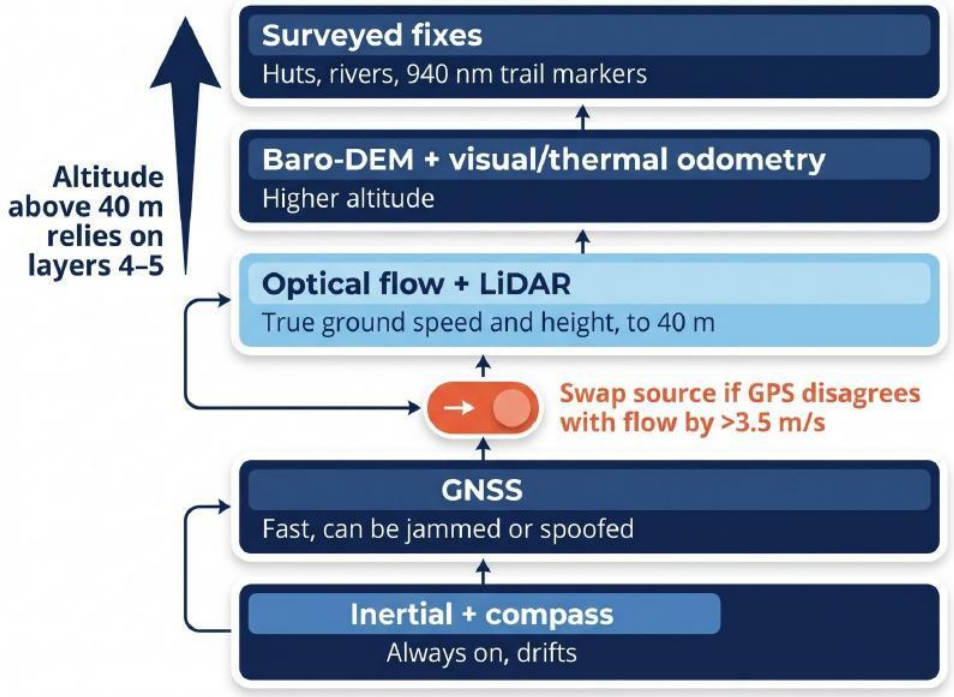
Prediction is not seeing through foliage.

Meet the Hiker Where They Will Be, or Search the Area Systematically



Timings and search geometry are modeled in simulation, not field-measured.

Navigation in Layers, So a Jammed GPS Does Not Mean a Lost Drone



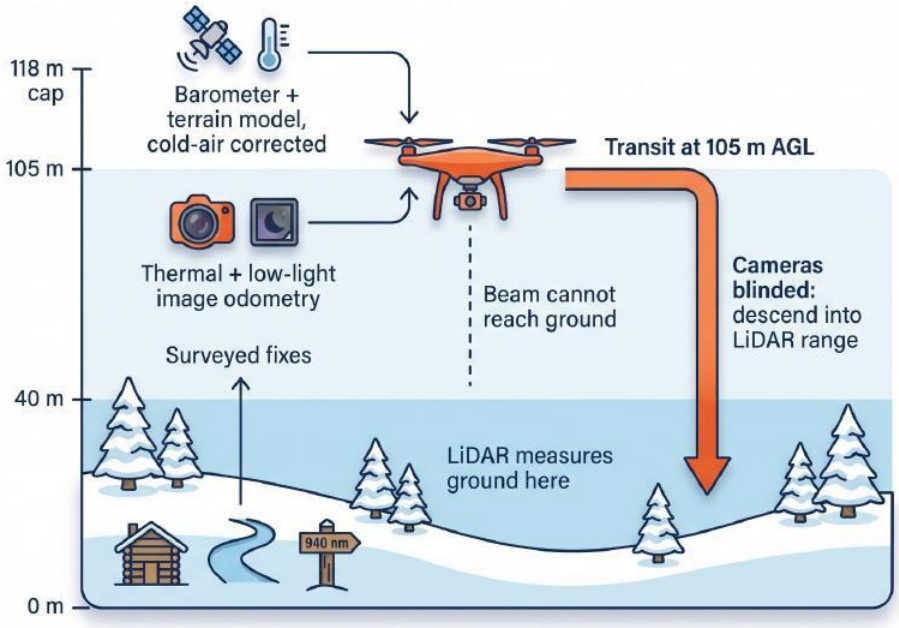
Design rules

ArduPilot EKF3 holds two source sets: **GNSS** or **optical flow + LiDAR**

- Jamming indicator triggers a pre-emptive switch
- Above 40 m, height comes from cold-corrected barometer and terrain model
- Surveyed landmarks reset drift

Qualification: source switching is tested in simulation only, not flight-proven. Navigation above 40 m: next slide.

Above 40 m the LiDAR Cannot See the Ground, So Other Layers Take Over



CONCEPT — SIMULATED

The gap

TF02-Pro LiDAR reference range is **40 m**

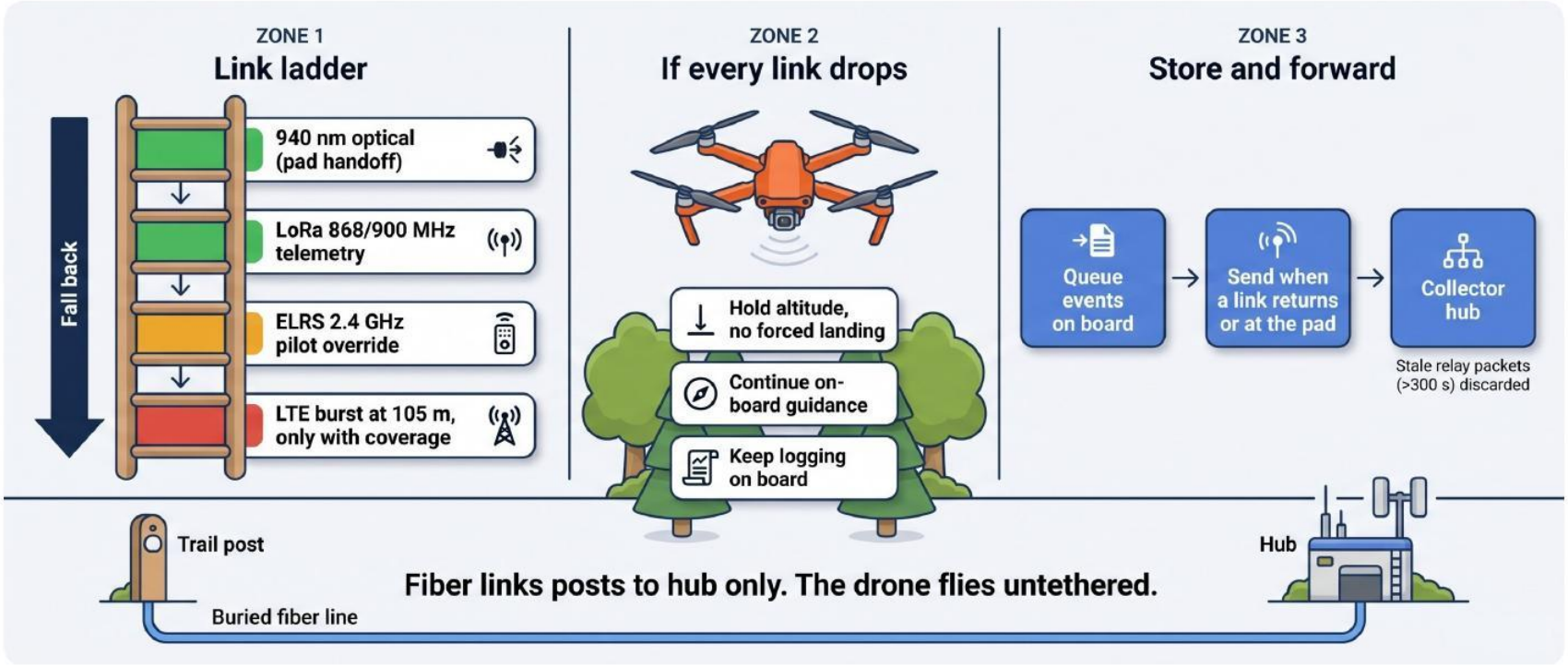
At **105 m** transit it cannot measure terrain

High-altitude layers (FIN-LAB-03 concept)

- Barometric height scaled for cold air, checked against a terrain model
- Image odometry from the thermal and Starvis cameras
- Watchdog for a frozen thermal shutter
- Drift reset over surveyed huts, rivers and trail-post markers

Qualification: simulated 30-min drift of **20.76 m** against a **30 m** pass mark. Simulation only, not flight data.

Four Links, a Lost-Link Plan and Store-and-Forward



Resilient by design, not jam-proof. Lost-link behaviour is simulated; store-and-forward is a design concept.

Once Found, a Human Pilot and a Human Voice Take Over

Why hand over

- Close-in obstacles in the dark need human judgement
- A frightened person trusts a named rescuer more than a silent robot

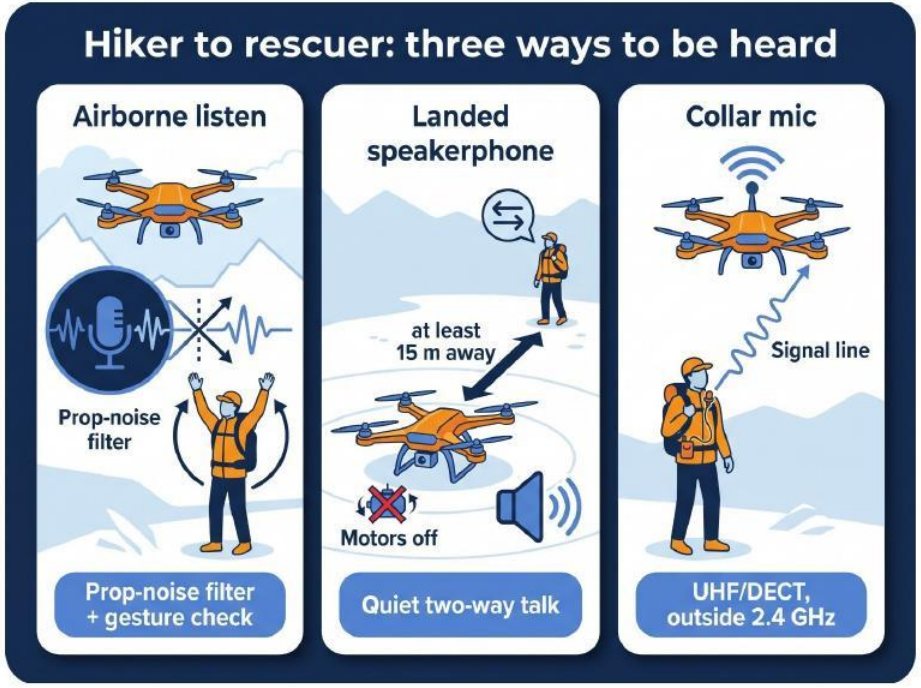
Pilot controls

Hold at **35 m AGL / 25 m** offset

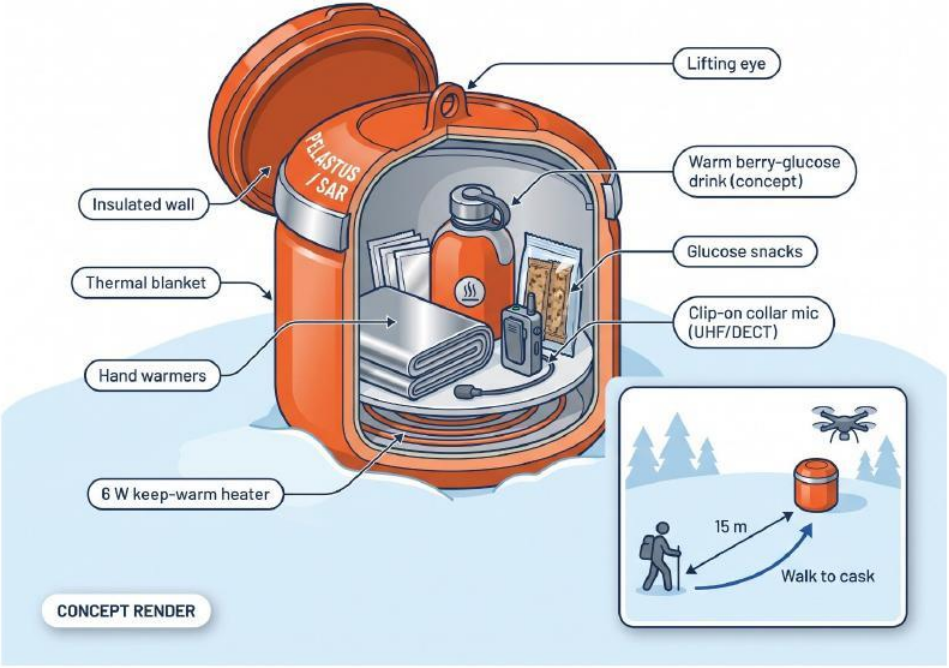
- Remote nudges or handheld RC (LOITER, GUIDED, RTL)
- 15 m standoff and 6 m LiDAR floor stay enforced

Three ways the hiker is heard: airborne listening, landed speakerphone, collar mic from the cask

Voice levels are modeled, not measured.



The Rescue Cask: Warmth, Comfort and a Voice Link



Delivery rules

Lowered onto clear snow **15 m** from the person, never overhead

25 N breakaway: a snagged line cannot pull the drone down

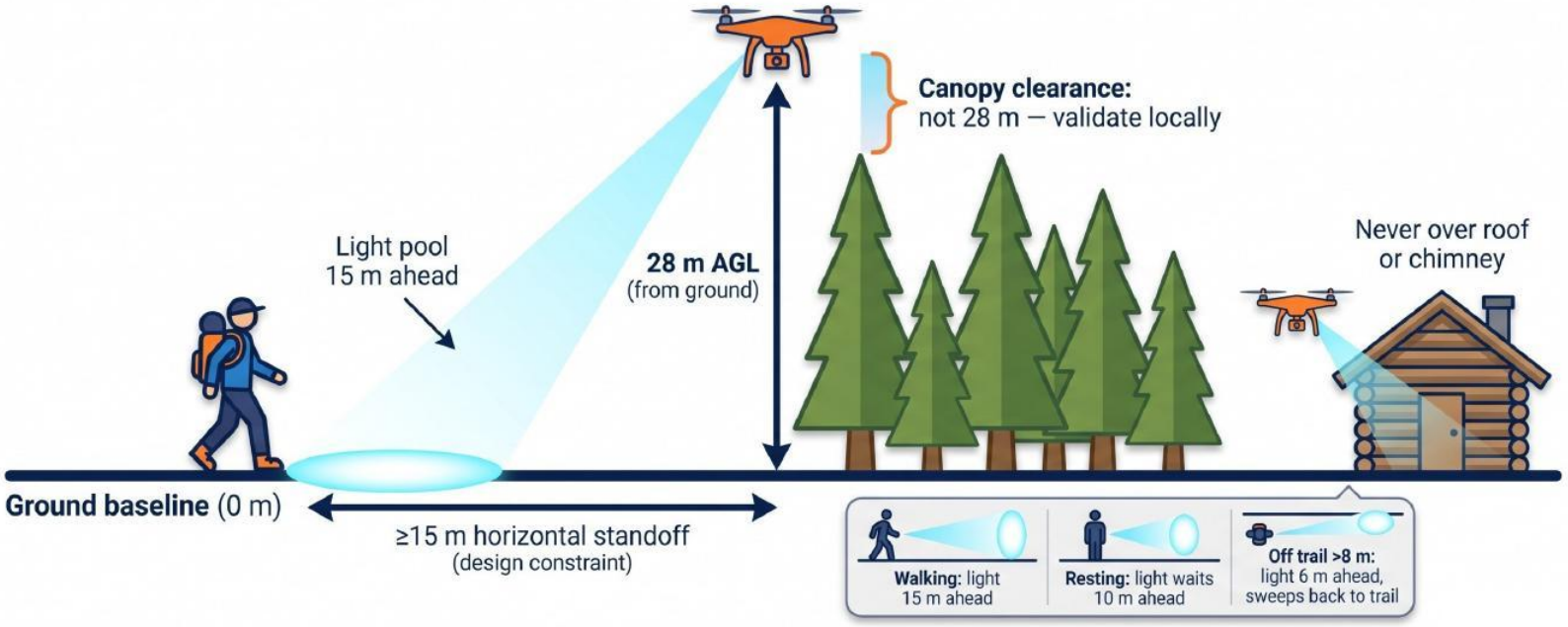
Drone climbs to **50 m** before it speaks

Public specification

- Comfort items only, alcohol-free
- Collar mic lets the hiker talk back

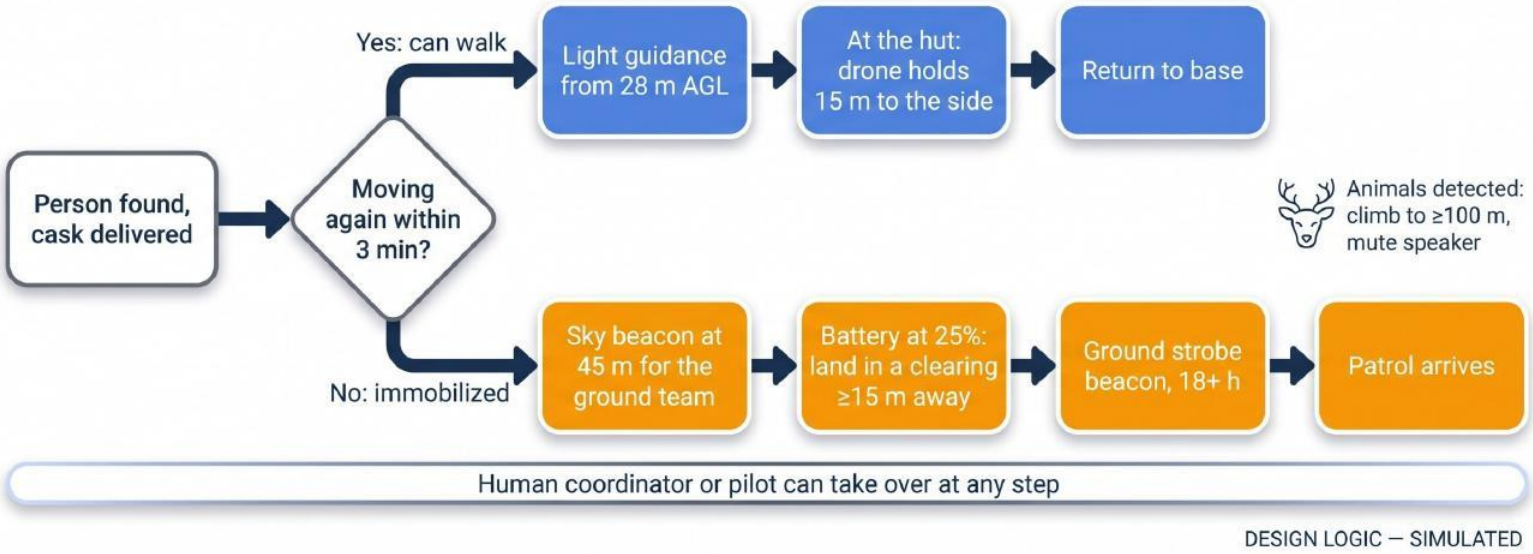
Qualification: design targets. No medical treatment instructions and no claim about survival time.

Opaslyhty: Leading the Hiker to the Hut With Light



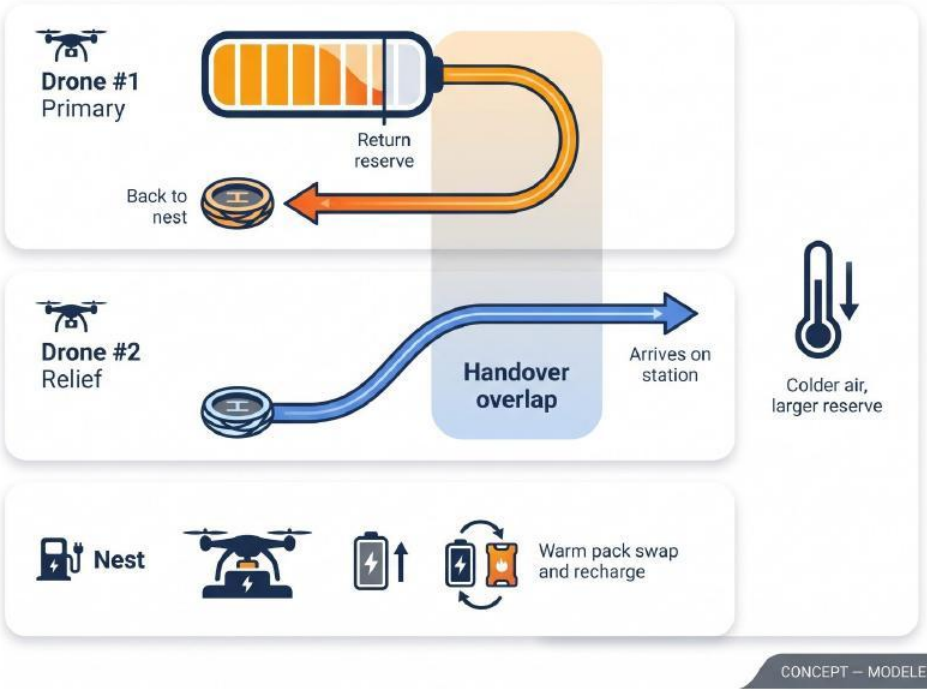
28 m AGL is measured from the ground, not from the tree crowns; canopy clearance must be validated locally. The ≥15 m horizontal standoff is a design constraint, not certified safety.

Two Branches After Contact: Walk to the Hut, or Shelter in Place



Thresholds are design values tested in simulation; the 18+ h beacon is a planning estimate.

Relief Before Reserve: Keeping a Drone Over the Hiker



CONCEPT — MODELED

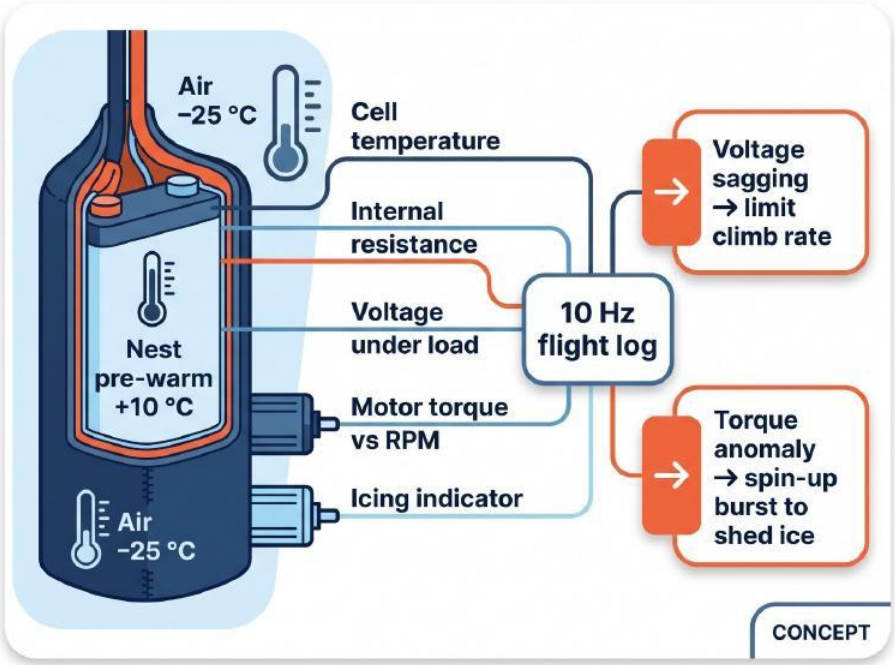
Air temperature	Return reserve
≥ 0 °C	30%
≤ −15 °C	40%
≤ −25 °C	45%

- #2 Backup is re-checked live before it is promoted
- Relief battery is checked before launch

Pre-warmed endurance **36–38 min** (planning estimate)

Qualification: reserve rules are design values; handover timing is modeled.

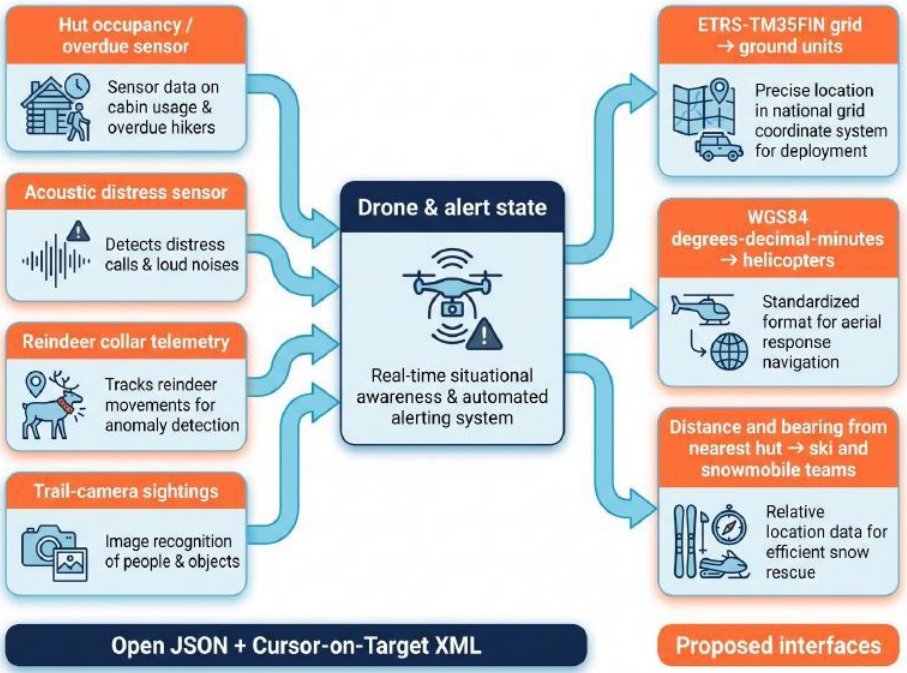
Kaamos Winter: Battery Care and a 10 Hz Black Box



Parameter	Planning value
Pack	8,000 mAh 4S2P, 118.4 Wh
Endurance, pre-warmed	36–38 min
Endurance, cold-soaked −25 °C	28+ min
Climb limit on sag	1.5 m/s below 12.2 V
Icing trigger	>12% torque anomaly
Shed burst	8,500 RPM

Qualification: planning estimates. Icing tolerance is unproven; the logger's grade label is a software classification, not laboratory qualification.

FIN-LAB Sensor Adapters Feed One Common Ingest



Adapters (FIN-LAB-01 Open Ingest API v0.9)

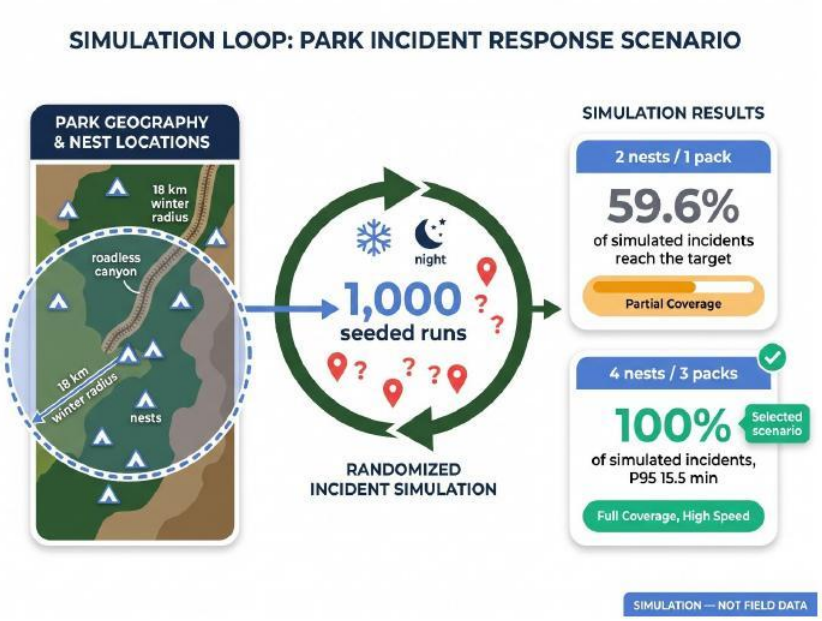
- Hut occupancy and overdue check-in, with a stove false-alarm guard
- Acoustic distress, with a geometry gate and wind suppression
- Reindeer-herding collar telemetry
- Trail-camera radar and optical sightings

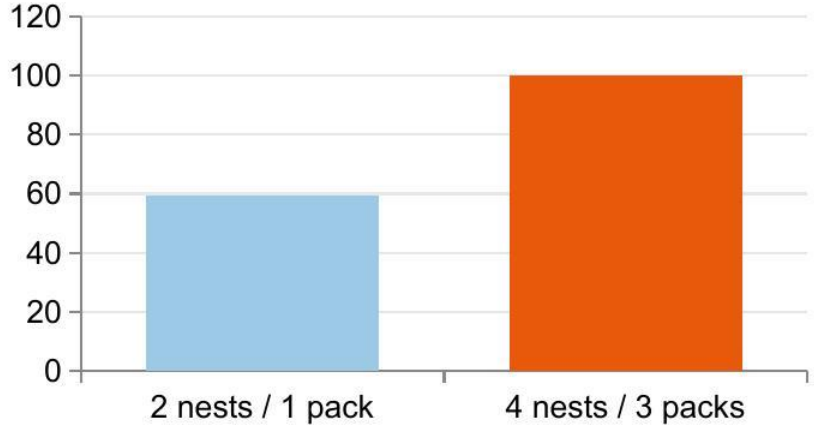
Common outputs

- ETRS-TM35FIN, WGS84 DDM, hut-relative bearing
- Open JSON + Cursor-on-Target XML, conformance scorecard

Qualification: proposed interfaces, simulated adapters. No live agency integration or partnership is claimed.

Sizing the Fleet With a Simulated Digital Twin





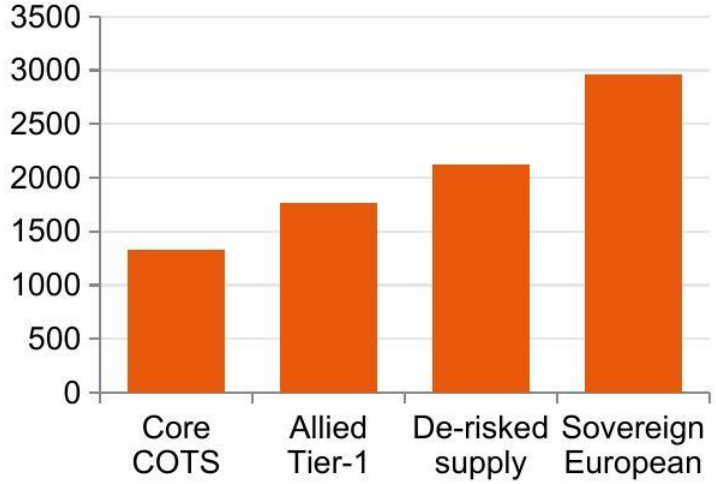
1,000-run seeded Monte Carlo (FIN-LAB-04)

Selected scenario: **4 nests, 3 packs, ~€185k**, P95 **15.5 min**

- *All figures are simulation outputs, not field results. Validate against real incident data.*

Eight BOM Lines and Four Sourcing Tiers

Core drone subsystem	EUR
Airframe, autopilot, GNSS, telemetry	490
Pi 5 companion and bench kit	115
Thermal + Starvis cameras	255
LiDAR + optical flow	90
Spotlight, sky-flare, release, speaker, cask	99
Arctic battery and charger	155
Safety-pilot RC and cells	80
EMI ferrites and collar mic	45
Total core drone	1,329

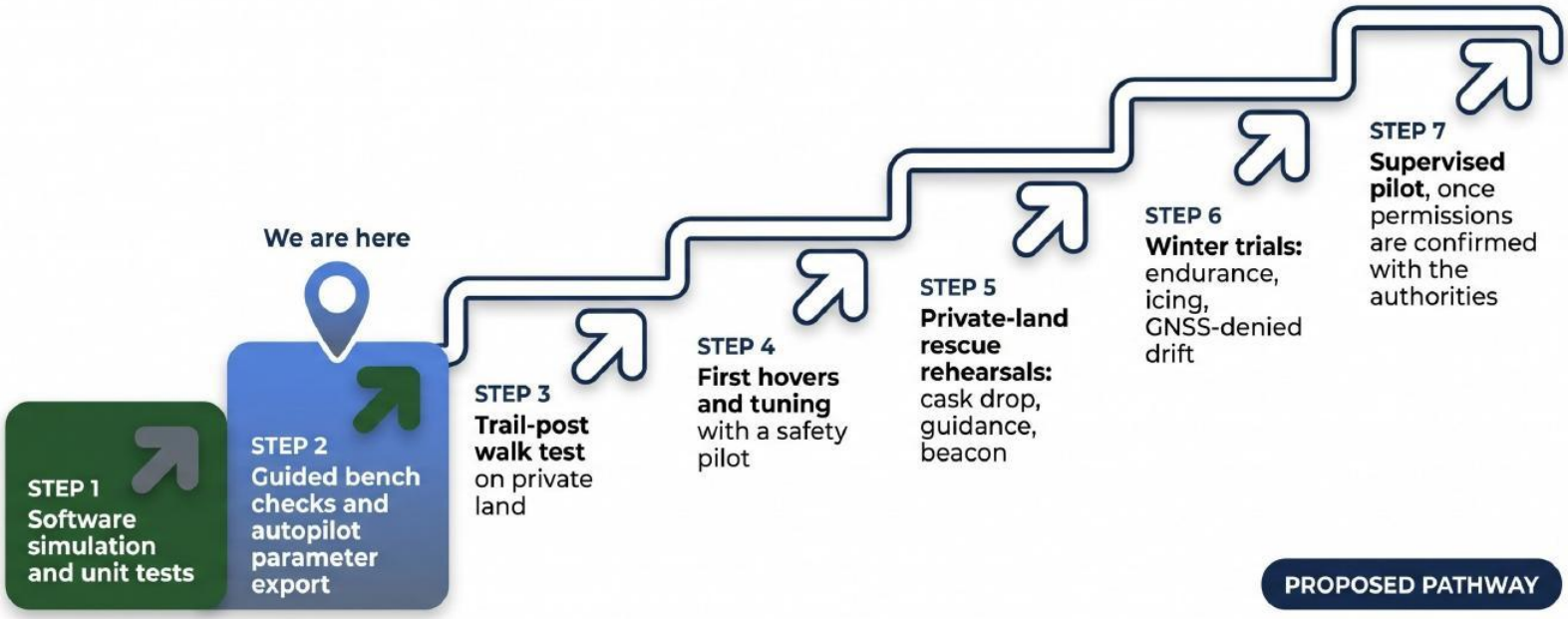


All tiers under the project's **€5,000** rule

Planning estimates, not quotations. Compatibility and procurement still to verify

Optional heated clamshell pad +€212. Trail posts, nests, deployment, training and programme costs are separate.

From Bench to Field: The Validation Ladder



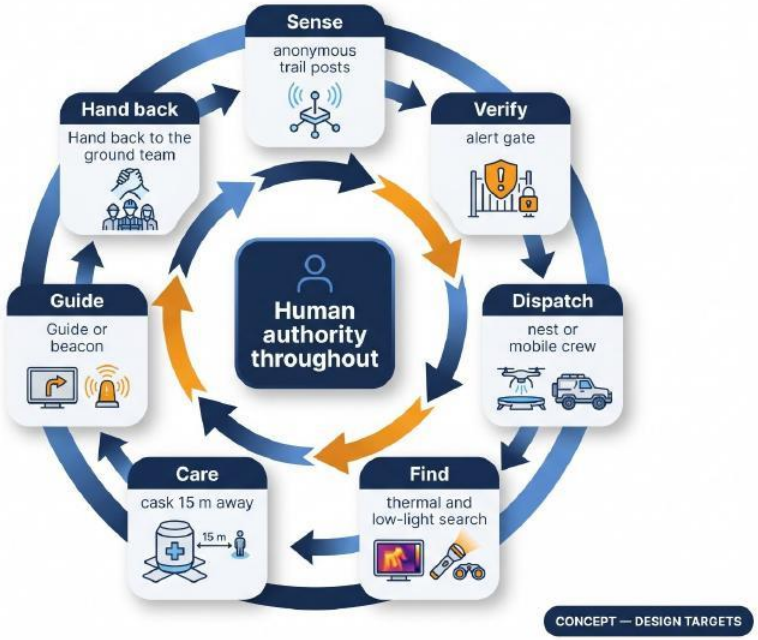
Proposed pathway built from the design log and Finnish brief. Steps 3–7 have not been completed. Operational permissions and pilot requirements must be confirmed with Traficom and the tasking authorities; no certification or legal conclusion is implied.

Readiness and Verification Matrix

Capability	Designed	Simulated / unit-tested	Hardware bench	Field-validated	Authorised
Trail post and 48 h buffer	Yes	Yes (log)	Partial	Not yet	Not yet
Alert gate, dispatch, ranking	Yes	Yes (log)	n/a	Not yet	Not yet
Navigation layers and high altitude	Yes	Yes (log)	Partial	Not yet	Not yet
Links and lost-link behaviour	Yes	Yes (log)	Partial	Not yet	Not yet
Cask drop and guidance geometry	Yes	Yes (log)	Not yet	Not yet	Not yet
Immobilized-hiker beacon	Yes	Yes (log)	Not yet	Not yet	Not yet
Human pilot and two-way voice	Yes	Yes (log)	Partial	Not yet	Not yet
Winter battery and black box	Yes	Yes (log)	Not yet	Not yet	Not yet
FIN-LAB adapters and ingest	Yes	Yes (log)	n/a	Not yet	Not yet
Fleet digital twin	Yes	Yes (log)	n/a	Not yet	n/a

29 unit-test suites reported passing in the design log (historical, not re-run) · audit findings **F01–F53** recorded as resolved · physical assembly and live tuning self-reported in the source as about 20% (historical, unverified). *"Partial" means bench tooling or parts exist but assembly is incomplete. "Authorised" requires confirmation by the relevant authorities. Not a certified aircraft or medical device.*

Recap and Sources



Source